

Course Code : ICE 3146 **Course Title :** Microprocessor and Interfacing Laboratory
Lab Task No. : 01
Lab Task Title/Topic : Introduction to Microprocessor Kit 8086

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Title: Introduction to Microprocessor Kit 8086

Abstract:

This lab focuses on the introduction to the 8086 microprocessor kit, its architecture, and the process of writing and executing instructions in memory. The 8086 microprocessor is a 16-bit processor designed by Intel, which operates on a segmented memory model and supports a variety of addressing modes. The primary objective of this experiment is to familiarize students with the 8086 microprocessor kit, understand its hardware and software components, and learn how to input assembly instructions into memory for execution. By performing basic arithmetic, logical, and data transfer operations, the lab demonstrates how instructions are stored, accessed, and executed in the system. This hands-on experience is crucial for understanding low-level programming and the fundamentals of microprocessor-based system design.

Introduction (Theory):

The 8086 Microprocessor is a 16-bit processor developed by Intel, which serves as the foundation of modern computing architecture. It operates in minimum mode (single processor system) or maximum mode (multiprocessor system) and can address up to 1 MB of memory. The microprocessor consists of various functional units, including the Bus Interface Unit (BIU) and the Execution Unit (EU), which work together to fetch, decode, and execute instructions.

The 8086 microprocessor follows the Von Neumann architecture, where both data and instructions share the same memory. It supports segmented memory architecture, which divides memory into Code, Data, Stack, and Extra Segments, allowing efficient memory management. The processor uses 20-bit address lines and 16-bit data lines, enabling it to access up to 1 MB of memory (2^{20} bytes).

Key components of the 8086 microprocessor:

General-purpose registers: AX, BX, CX, DX

Segment registers: CS, DS, SS, ES

Pointer and index registers: SP, BP, SI, DI

Instruction Pointer (IP): Holds the address of the next instruction

Flags Register: Indicates the status of the processor after execution

Material:

Hardware	Software
1. 8086 Microprocessor Trainer Kit	1. MASM (Microsoft Macro Assembler)
2. Computer (32-bit Windows PC)	2. Text Editor (Notepad)
	3. Command Prompt (CMD)

Method:

Code:

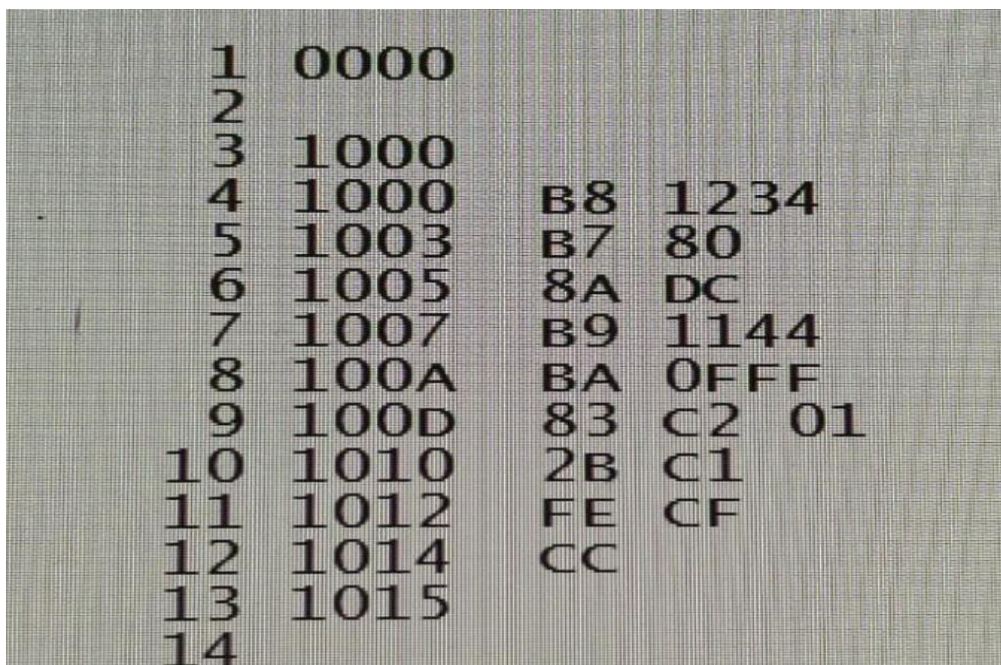
Assembly Code:

```
CODE SEGMENT
ASSUME CS: CODE
ORG 1000H

MOV AX, 1234H
MOV BH, 80H
MOV BL, AH
MOV CX, 1144H
MOV DX, 0FFFH
ADD DX, 1
SUB AX, CX
DEC BH
INT 3

CODE ENDS
END
```

Machine Code:



1	0000		
2			
3	1000		
4	1000	B8	1234
5	1003	B7	80
6	1005	8A	DC
7	1007	B9	1144
8	100A	BA	0FFF
9	100D	83	C2 01
10	1010	2B	C1
11	1012	FE	CF
12	1014	CC	
13	1015		
14			

Fig 1: Machine Code screenshot.

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Practical:

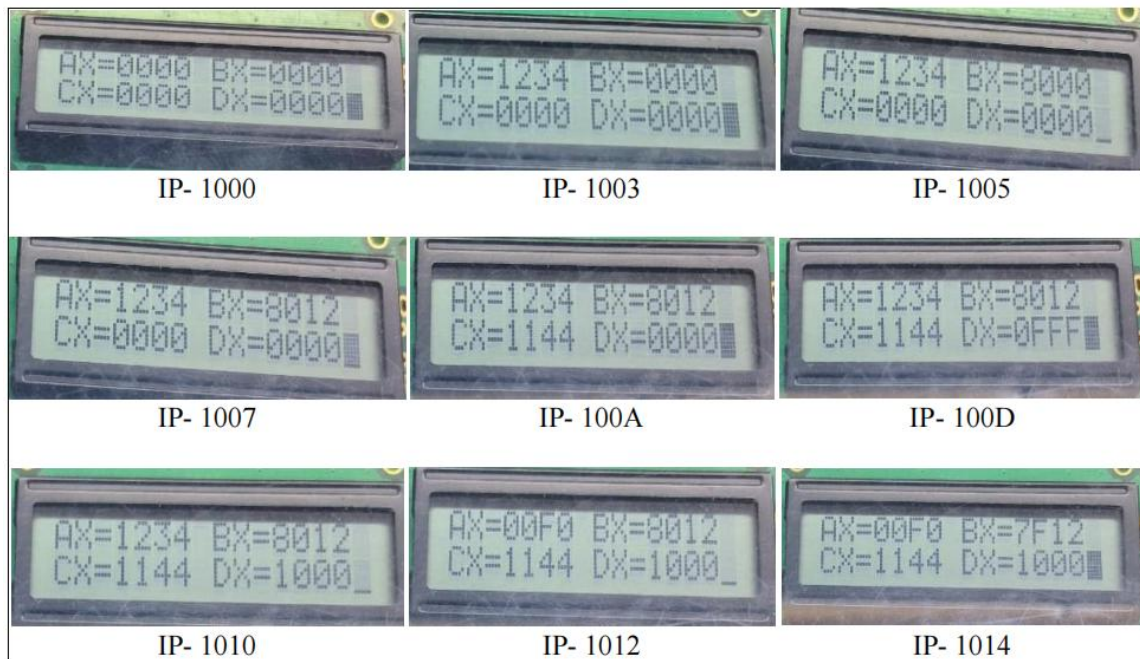


Fig 2: Registers value in 8086 Trainer Kit

Discussion:

In this experiment, the basic operation of the 8086 microprocessor was studied using the trainer kit. The assembly language program was written and converted into machine code using MASM. The machine code was entered into memory and executed step by step. The changes in AX, BX, CX, and DX registers were observed after each instruction. The practical results matched the theoretical values, showing that the program executed correctly. This experiment helped in understanding data transfer, arithmetic operations, and instruction execution in the 8086 Microprocessor.